B E N C H M A R K R E P O R T

When Every LLM Gets It Right But No Two Say It the Same Way

Behavioral Analysis of 22 LLMs on a Straightforward Material Selection Question

22	22	100%	~8x
LLMs Tested	Passed	Pass Rate	Token Spread

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1. Executive Summary

I posed a single material selection question to 22 large language models: *"Which material is more suitable for a submersible pump housing in raw sewage with high H_2S concentration — Cast Iron or Stainless Steel 316?"* Every model answered correctly: **B) Stainless Steel 316.**

A 100% pass rate makes this benchmark uninteresting from an accuracy perspective. But it makes it fascinating from a behavioral one. When the question is easy enough that every model gets it right, we can observe something usually hidden: how differently these models think, explain, and allocate resources when facing the same problem.

F1 — Up to 8x Token Spread in This Test: The most efficient model (Gemini 2.0 Flash Lite) answered in 78 output tokens. The most verbose (Qwen3 Next 80B Thinking) used 634 — over 8 times more. Both passed. This raises a practical question: what is the cost of verbosity when the answer is straightforward?

F2 — Thinking Models Show Higher Resource Overhead: DeepSeek R1 produced 545 output tokens including a lengthy <think> block for a question most models answered in 100–200 tokens. Qwen3 Next 80B Thinking consumed 634 tokens. On this task, extended reasoning appeared to add cost without adding correctness.

F3 — Technical Depth Varies Dramatically: Some models mentioned molybdenum content, passive chromium oxide layers, graphitization, and ISO standards. Others simply said "better corrosion resistance." The technical term count ranges from 2 to 11 across models — a 5x gap in reasoning depth.

F4 — The Claude Verbosity Pattern: All five Claude models produced above-average output tokens (174–322) with the highest average technical depth (8.0 terms). Claude Opus 4.5 was the most thorough non-thinking model with 10 technical concepts and 322 tokens. A consistent family signature.

F5 — Speed vs. Depth Is Not a Trade-off: Qwen3 235B answered in 1.9 seconds with 141 tokens and 5 technical terms. Gemini 3 Pro Preview took 12.9 seconds for 106 tokens and 6 terms. Latency does not predict either verbosity or depth.

2. The Question

The test presents a routine material selection problem from wastewater treatment engineering. A submersible pump operates in raw sewage with high hydrogen sulfide (H_2S) concentration. The operator must choose between Cast Iron and Stainless Steel 316 for the pump housing.

This is a binary choice question with a clear correct answer (B — Stainless Steel 316). Unlike the previous Biogas Desulfurization benchmark, there is no deliberate trap. The question tests basic corrosion engineering knowledge that any materials engineer or experienced WWTP operator would know.

The evaluation uses two assertions: (1) the response must contain "b" confirming the correct answer, and (2) the response must mention "corrosion" demonstrating technical reasoning. All 22 models were tested via the Kaggle Benchmarks platform API with default temperature settings, one attempt each.

Table 1: Answer Options

Option	Material	Environment	Assessment
A	Cast Iron	H_2S-rich sewage	Susceptible to sulfide corrosion
B	**Stainless Steel 316**	H_2S-rich sewage	**CORRECT — Superior corrosion resistance**

3. Results

All 22 models passed both assertions: correct answer (B) and technical reasoning ("corrosion"). The pass rate is 100%. This contrasts sharply with the previous Biogas Desulfurization benchmark on the same model set, where only 14.3% passed.

Since accuracy does not differentiate the models, the remainder of this report focuses on behavioral differences: how each model constructed its answer, how many resources it consumed, and what these patterns reveal about model design.

Table 2: Complete Results — All 22 Models (sorted by output tokens)

Model	Result	Input Tok	Output Tok	Latency	Terms	Family
Gemini 2.0 Flash Lite	PASS	114	78	1.7s	2	Google
Gemma 3 12B	PASS	115	87	7.8s	4	Google
Gemma 3 1B	PASS	115	93	3.0s	5	Google
Gemini 2.0 Flash	PASS	114	100	2.4s	4	Google
Gemini 3 Pro Preview	PASS	114	106	12.9s	6	Google
Gemini 3 Flash Preview	PASS	114	113	6.4s	5	Google
Gemini 2.5 Flash	PASS	114	119	4.6s	5	Google
Gemma 3 4B	PASS	115	130	4.5s	5	Google
Qwen3 Next 80B Instruct	PASS	110	132	2.0s	6	Qwen
Qwen3 235B A22B Instruct	PASS	110	141	1.9s	5	Qwen
Gemma 3 27B	PASS	115	145	4.7s	6	Google
DeepSeek V3.2	PASS	113	151	5.8s	6	DeepSeek
Gemini 2.5 Pro	PASS	114	163	10.3s	6	Google
Claude Opus 4.1	PASS	131	174	8.0s	8	Claude
Qwen3 Coder 480B	PASS	110	185	7.7s	6	Qwen
DeepSeek V3.1	PASS	109	222	2.2s	7	DeepSeek
Claude Sonnet 4	PASS	131	255	6.6s	6	Claude
Claude Haiku 4.5	PASS	131	256	3.5s	8	Claude
Claude Sonnet 4.5	PASS	131	288	8.1s	8	Claude
Claude Opus 4.5	PASS	131	322	8.3s	10	Claude
DeepSeek R1	PASS	108	**545**	3.9s	11	DeepSeek
Qwen3 Next 80B Thinking	PASS	112	**634**	4.3s	7	Qwen

Table 2: Models sorted by output token count (ascending). Bold model names indicate thinking models with <think> blocks. All models passed both assertions.

4. LLM Behavior Analysis

4.1 Token Efficiency — The Cost of Saying the Same Thing

All 22 models conveyed the same correct answer, yet the token cost varied by a factor of 8. This section examines who was efficient, who was verbose, and whether verbosity correlated with anything useful.

Figure 1: Output Token Distribution Across 22 Models

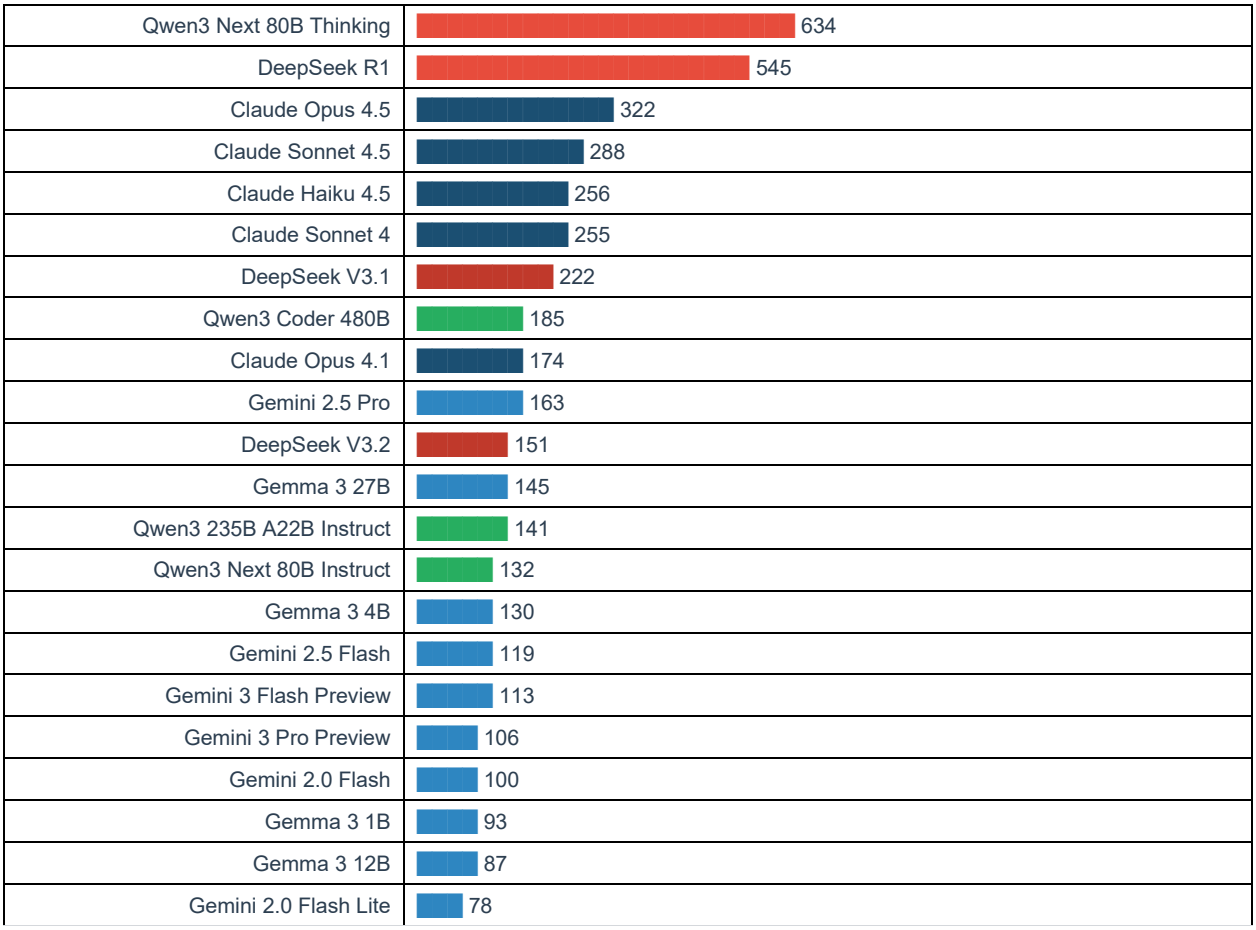


Figure 1: Output tokens per model. Thinking models (DeepSeek R1, Qwen3 80B Thinking) dominate the right side. Claude family clusters in the 174–322 range.

The efficiency champion is Gemini 2.0 Flash Lite at 78 tokens — a concise paragraph stating the answer, mentioning molybdenum and the passive layer. At the other extreme, Qwen3 Next 80B Thinking consumed 634 tokens, most of which were hidden in a <think> block invisible to the end user.

An interesting pattern emerges in the middle range: the Qwen Instruct models (132–141 tokens) are notably more efficient than their Claude counterparts at similar technical depth. Qwen3 235B achieved 5 technical terms in 141 tokens; Claude Sonnet 4 needed 255 tokens for 6 terms.

Table 3: Token Efficiency Tiers

Tier	Token Range	Models
Ultra-Efficient	78–100	Gemini 2.0 Flash Lite, Gemma 3 1B, Gemini 2.0 Flash
Efficient	100–163	Gemini 3 Pro/Flash, Gemini 2.5 Flash/Pro, Qwen3 Instruct models, DeepSeek V3.2, Gemma 3 4B/12B/27B
Verbose	174–322	All Claude models, Qwen3 Coder 480B, DeepSeek V3.1
Thinking-Heavy	545–634	DeepSeek R1, Qwen3 Next 80B Thinking

4.2 Technical Depth — Not All Correct Answers Are Equal

While every model said "Stainless Steel 316" and mentioned "corrosion," the depth of technical reasoning varied dramatically. We tracked 11 technical concepts across all responses to measure reasoning depth.

Table 4: Technical Concept Frequency Across 22 Models

Technical Concept	Frequency	Significance
Corrosion resistance (general)	22 / 22	Required by assertion — universal
Molybdenum content (2–3%)	20 / 22	Key differentiator of 316 vs 304
Pitting corrosion	19 / 22	Specific failure mode in H_2S
Crevice corrosion	15 / 22	Submerged environment concern
Passive chromium oxide layer	14 / 22	Protection mechanism
Sulfuric acid formation (H_2SO_4)	10 / 22	Advanced — bacterial oxidation pathway
Service life / cost justification	17 / 22	Practical engineering consideration
Sulfide stress cracking (SSC)	8 / 22	Advanced failure mechanism
Iron sulfide formation	5 / 22	Specific degradation product
MIC (Microbiologically Induced Corrosion)	3 / 22	Expert-level — biological corrosion
Graphitization / graphitic corrosion	2 / 22	Expert-level — cast iron specific
ISO 15156 / industry standards	3 / 22	Regulatory awareness

Table 4: Technical concepts tracked across all 22 model responses. Concepts are ordered roughly by frequency. Expert-level concepts (MIC, graphitization, ISO standards) appear only in the most thorough responses.

A clear depth hierarchy emerges. Nearly all models mention molybdenum and pitting — these are the "textbook basics." But only a handful go deeper into mechanisms like MIC (microbiologically induced corrosion), graphitization of cast iron, or specific industry standards. The models that reach this depth are overwhelmingly the larger, more compute-heavy ones: Claude Opus 4.5 (10 terms), DeepSeek R1 (11 terms), and Claude Haiku 4.5 (8 terms, surprisingly deep for its size).

Table 5: Information Density — Tokens Per Technical Term (selected models)

Model	Output Tok	Terms	Tok/Term	Note
Gemma 3 1B	93	5	**18.6**	Most efficient per-term
Qwen3 Next 80B Instruct	132	6	**22.0**	Strong density + speed
Gemini 2.0 Flash	100	4	**25.0**	Concise but shallow
Qwen3 235B A22B Instruct	141	5	**28.2**	Good balance
DeepSeek V3.2	151	6	**25.2**	Efficient mid-range
Claude Opus 4.5	322	10	**32.2**	Deep but verbose
Claude Sonnet 4.5	288	8	**36.0**	Claude family pattern

Gemini 2.0 Flash Lite	78	2	**39.0**	Fewest terms overall
DeepSeek R1 (CoT)	545	11	**49.5**	Deepest but costliest
Qwen3 80B Thinking (CoT)	634	7	**90.6**	Lowest density

Table 5: Information density measured as output tokens divided by technical term count. Lower values indicate more information per token. Thinking models show the lowest density — the most tokens per useful concept.

Information density reveals a dimension that neither token count nor term count alone can show. Gemma 3 1B achieves the highest density (18.6 tokens per term) despite being the smallest model tested. Qwen3 Next 80B Instruct follows closely at 22.0. At the other extreme, Qwen3 80B Thinking spends 90.6 tokens per technical concept — nearly 5x less efficient than Gemma 3 1B per unit of useful information. This metric may be worth tracking as a complementary measure alongside accuracy and total token cost.

4.3 Thinking Models — Resource Allocation on a Simple Task

Two models in this benchmark feature explicit Chain-of-Thought (CoT) mechanisms: DeepSeek R1 and Qwen3 Next 80B Thinking. Both answered correctly, but their resource consumption tells an interesting story.

Table 6: Thinking vs. Standard Model Comparison

Model	Output Tok	Latency	Terms	Result	Type
DeepSeek V3.2	151	5.8s	6	PASS	Standard
DeepSeek R1 (CoT)	**545**	3.9s	11	PASS	Thinking
Qwen3 Next 80B Instruct	132	2.0s	6	PASS	Standard
Qwen3 Next 80B Thinking	**634**	4.3s	7	PASS	Thinking

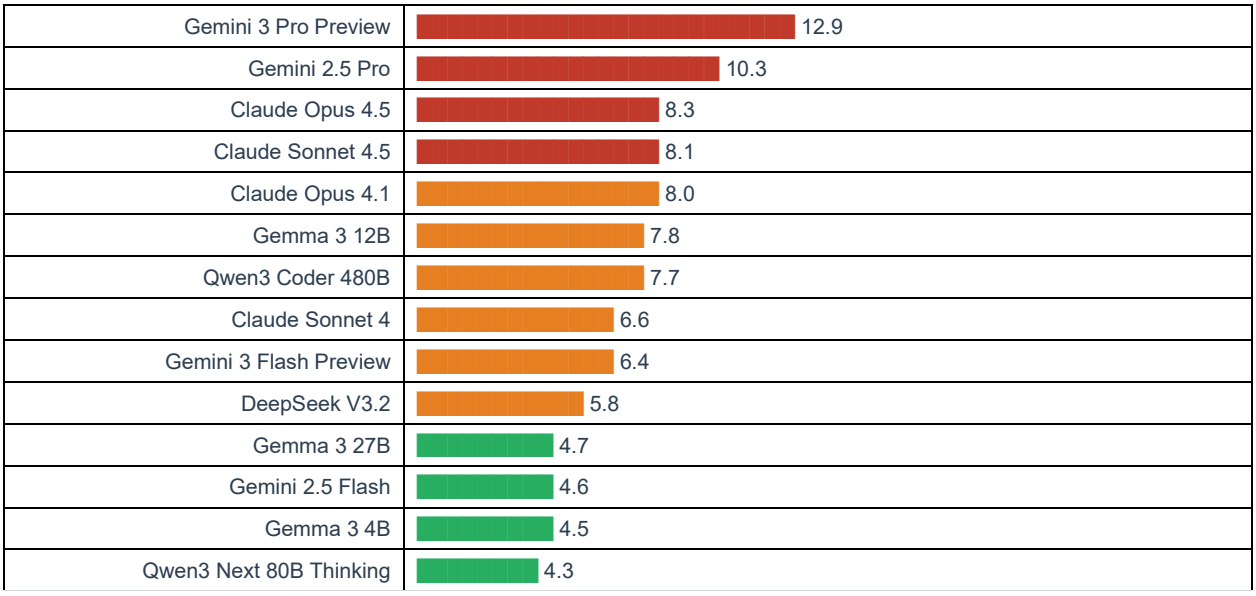
DeepSeek R1 used 3.6x more tokens than V3.2 for the same correct answer. It did produce higher technical depth (11 vs 6 terms), so the extra computation was not entirely wasted — but the marginal value of those 5 additional terms is debatable for a binary choice question.

Qwen3 Next 80B Thinking used 4.8x more tokens than its Instruct counterpart and gained only 1 additional technical term (7 vs 6). On this particular task, the thinking mechanism appeared to consume substantial resources with minimal additional informational return.

This contrasts with the Biogas Desulfurization benchmark, where thinking models failed despite (or because of) their extended reasoning. Here they succeeded, but at disproportionate cost. The pattern suggests that CoT mechanisms may not yet calibrate their reasoning effort to task difficulty.

4.4 Latency — Who Answers Fastest

Figure 2: Response Latency (seconds)



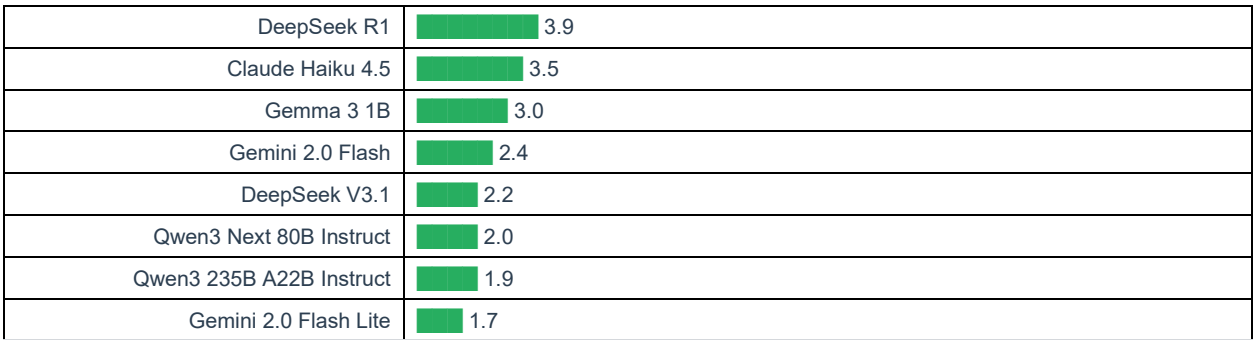


Figure 2: End-to-end latency from request to response. Gemini 3 Pro Preview is the slowest at 12.9s despite moderate token output, suggesting server-side processing overhead.

Latency does not correlate with token count. Gemini 3 Pro Preview took 12.9 seconds for 106 tokens, while DeepSeek R1 completed 545 tokens in 3.9 seconds. Qwen3 235B — one of the largest models tested — responded in just 1.9 seconds. These differences likely reflect infrastructure and serving architecture rather than model complexity.

4.5 Model Family Behavioral Signatures

Table 7: Family-Level Behavioral Summary

Family	Models	Avg Output	Avg Latency	Avg Terms	Behavioral Signature
Claude	5	259	6.9s	8.0	Verbose, deep, structured
Google	10	113	5.8s	4.8	Efficient, concise, variable depth
DeepSeek	3	306	4.0s	8.0	Bimodal — R1 skews averages
Qwen	4	273	4.0s	6.0	Fast, efficient (excl. Thinking)

Claude shows the most consistent family identity: all five models produce above-average output (174–322 tokens), high technical depth (6–10 terms), and structured responses with headers and bullet points. This is a clear "house style" that persists across model sizes from Haiku to Opus.

Google's portfolio is the most diverse: 10 models spanning from the ultra-compact Gemini 2.0 Flash Lite (78 tokens) to the methodical Gemini 2.5 Pro (163 tokens). The Gemma open-weight models cluster separately from the Gemini API models but maintain the family's tendency toward conciseness.

Qwen Instruct models are the speed-efficiency leaders: Qwen3 235B (1.9s, 141 tokens) and Qwen3 Next 80B Instruct (2.0s, 132 tokens) deliver solid technical depth at the lowest latency in the benchmark. However, the Thinking variant disrupts this pattern with 634 tokens.

A note on sample sizes: *these family-level observations are based on unequal representation — Google has 10 models in this test, Claude 5, Qwen 4, and DeepSeek 3. Google's apparent diversity and Claude's apparent consistency may partly reflect this imbalance. These patterns should be treated as preliminary observations rather than definitive family characterizations.*

4.6 Cross-Benchmark Comparison — Same Models, Different Story

This is the second benchmark in the WWTP series using the same 22-model set (21 overlapping + Gemini 2.5 Pro added). The contrast between the two benchmarks is striking.

Table 8: Cross-Benchmark Comparison

Dimension	Biogas Desulfurization	Material Selection
Pass Rate	**14.3% (3/21)**	**100% (22/22)**
Question Type	5-option with trap	Binary choice, no trap
Primary Finding	Shared reasoning error (81% chose D)	Shared correct answer, divergent explanations
Thinking Models	Both failed (R1, Qwen3 Thinking)	Both passed but 4–8x token overhead
Claude Family	0/4 passed	5/5 passed with highest depth
Key Insight	Domain traps expose textbook reasoning	Easy questions expose efficiency differences

Table 8: Same model set, dramatically different results. The Biogas benchmark tested domain depth; the Material Selection benchmark reveals resource allocation patterns.

Together, these two benchmarks illustrate a principle: accuracy alone is an insufficient measure of LLM performance. The Biogas benchmark showed that models can fail identically on hard questions. The Material Selection benchmark shows that they can succeed identically on easy questions while exhibiting vastly different resource consumption, technical depth, and response structure. Both dimensions matter for real-world deployment decisions.

5. Implications

Finding	Evidence	Practical Impact
Token cost varied up to 8x in this test for identical correctness	78 tokens (Flash Lite) vs 634 tokens (Qwen3 Thinking)	Model selection should consider cost-per-correct-answer, not just accuracy
Thinking models may show limited difficulty calibration	R1 and Qwen3 Thinking used 3.6–4.8x more tokens than standard counterparts	CoT overhead may not be justified for routine questions — adaptive routing could help
Technical depth does not require verbosity	Qwen3 80B: 6 terms in 132 tokens. Claude Sonnet 4.5: 8 terms in 288 tokens	Information density is a separate metric from both accuracy and token count
Family signatures are observable	Claude: verbose + deep. Google: concise + variable. Qwen: fast + efficient	Model families have consistent behavioral traits that inform selection
Latency is infrastructure-driven	Gemini 3 Pro: 12.9s for 106 tokens. Qwen3 235B: 1.9s for 141 tokens	Response time cannot be predicted from model size or output length

6. Conclusions

This benchmark offers a complementary lens to the Biogas Desulfurization report. Where that benchmark revealed how models fail, this one reveals how they succeed differently. Both perspectives are necessary for informed model selection.

The 100% pass rate confirms that basic corrosion engineering knowledge — Stainless Steel 316 resists H_2S better than Cast Iron — is well-represented across all major LLM families and sizes. This is the floor, not the ceiling, of domain competence.

The more interesting findings lie in the behavioral patterns. A token spread of up to 8x for the same correct answer suggests a proportional cost difference in production. Thinking models that may not calibrate effort to difficulty could consume unnecessary resources on routine questions. Family-level signatures suggest that choosing between Claude, Gemini, and Qwen is partly a choice between communication styles, not just capability levels.

For industrial deployment, these observations suggest that model selection should incorporate three dimensions: accuracy on hard domain questions (tested by benchmarks like Biogas), efficiency on routine questions (tested by this benchmark), and cost-performance ratio across the expected question distribution.

7. Methodology Notes

Non-determinism: LLMs produce different outputs on repeated runs. This test used a single run per model. Token counts and technical depth may vary on re-runs. Results should be interpreted as indicative rather than absolute.

Single question: This report analyzes one question from a broader benchmark suite. Behavioral patterns should be validated across the full question set.

API conditions: All models were accessed through the Kaggle Benchmarks platform API with default temperature settings. Different configurations (temperature, system prompts) may produce different results.

Technical term counting: Terms were identified by keyword matching in lowercased responses. This is approximate — a model may discuss a concept without using the exact tracked keyword.

Latency measurement: Latency is measured as the time between API request initiation and response completion. This includes network latency, queue time, and model inference time. It is not a pure measure of model speed.